

Cheng Zhang

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PROFESSIONAL EXPERIENCE

X Square Robot

Jun. 2026 – Aug. 2026

Robot Learning Intern, Post-Training

Shenzhen, China

- Adapted $\pi 0.5$ and WallX VLAs to dual-arm, high-DoF dexterous robots, fine-tuning policies for hair-drying and fan-boxing with SE(3) relative EE control and language-conditioned multi-stage datasets
- Scaled VLA training and deployment across multi-node GPU clusters with JAX/FSDP, serving checkpoints as WebSocket policy servers and incorporating training-time Real-Time Chunking (RTC) to account for latency
- Improved 22-DoF dexterous hand tracking by deploying a real-time feed-forward residual network, reducing joint-tracking MAE by 20%; automated daily calibration and optimized inference for 120+ Hz control
- Developed Tac-Flow, a many-to-many tactile alignment framework mapping human glove data to multiple robot hands via conditional flow matching and cross-embodiment pairing
- Built Wuji-test and Sharpa-test diagnostic packages for scalable data collection, enabling non-robotics teleoperators to independently debug and recover hand, camera, and teleoperation failures
- Presented the dexterous hand demo at the 2026 World Robot Conference as team representative

RESEARCH EXPERIENCE

DITTO: Dexterous Interface for Transparent TeleOperation

May 2025 – Present

Research Assistant, ROAM Lab, Columbia University

New York, NY

- Architected a real-time force-feedback teleoperation system for a 7-DoF dexterous hand in ROS2 using Dynamixel actuators, achieving stable 150+ Hz low-latency closed-loop control
- Re-engineered 10-DoF hand kinematics in MuJoCo using CPU-based CEM optimization, achieving 96% workspace pose success ($3.7\times$ DexUMI and $3.0\times$ DOGlove) across five manipulation tasks
- Built a deterministic UR5 control framework with Slerp interpolation from 20 Hz policy inference to 125 Hz RTDE execution and Pinocchio-based collision checking; adopted across four lab projects to reduce control jitter and standardize policy evaluation
- Trained vision-conditioned Diffusion Policy on DITTO demonstrations for dexterous manipulation, achieving 90% success on contact-rich test-tube uncapping and validating policies on unimanual and bimanual tasks

MiniBEE: A New Form Factor for Compact Bimanual Dexterity

Nov. 2024 – Apr. 2025

Research Assistant, ROAM Lab, Columbia University

New York, NY

- Co-designed arm kinematics and developed a KD metric to evaluate candidate designs based on IK feasibility and Jacobian singularity checks
- Built an automated MoveIt benchmarking pipeline with custom RViz visualization, evaluating 6/7/8-DoF kinematic configurations over a 10×10 Cartesian workspace grid
- Quantified KD across 6/7/8-DoF kinematic configurations (2.7%, 68%, 98%), with the 8-DoF design preserving comparable workspace coverage versus the 12-DoF commercial ALOHA baseline
- Conducted moment-arm torque analysis to validate Dynamixel feasibility, guiding design changes that reduced estimated failure poses by 30% for real-world diffusion-policy experiments

EDUCATION

Columbia University

Jun. 2027 (Expected)

Bachelor of Science in Computer Science

New York, NY

- GPA: 4.0/4.0
- Relevant Coursework: Computational Aspects of Robotics, Deep Learning for Robot Manipulation, Robot Learning, Advances in Visual Generative Models, Machine Learning

PUBLICATIONS

- *DITTO: Dexterous Interface for Transparent TeleOperation* (Under Review, ICRA 2027)
- *MiniBEE: A New Form Factor for Compact Bimanual Dexterity* (Accepted, ICRA 2026)

ADDITIONAL INFORMATION

Technical Skills: Python, C++ | PyTorch, OpenCV | ROS2, MoveIt, MuJoCo, Pinocchio | Docker, GCP, AliCloud

Languages: English (proficient), Mandarin Chinese (native)

Interests: badminton, cello, traveling, running